STAT508 - HW3

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Chapter 7

Problem 9

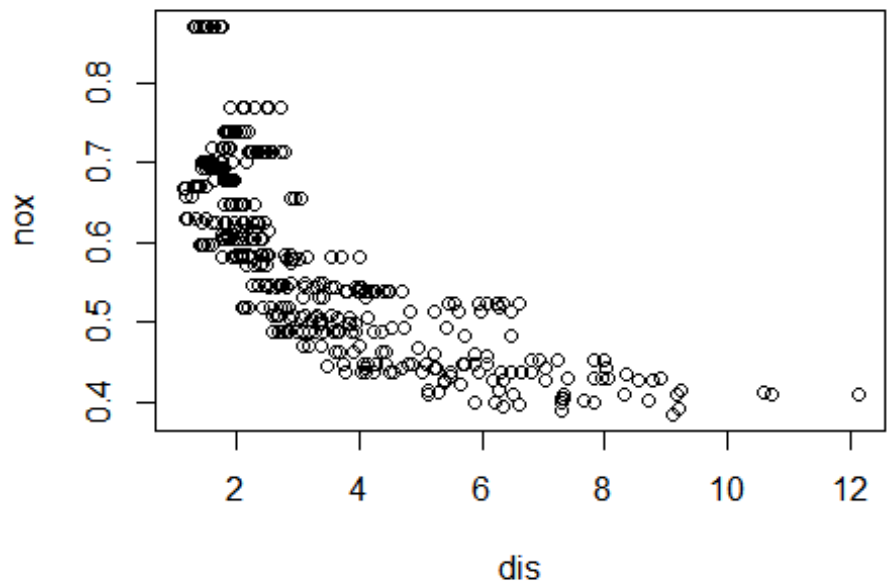
This question uses the variables dis (the weighted mean of distances to five Boston employment centers) and nox (nitrogen oxides concentration in parts per 10 million) from the Boston data. We will treat dis as the predictor and nox as the response

```
library(ISLR2)
data("Boston")
names(Boston)

##  [1] "crim"    "zn"      "indus"   "chas"    "nox"     "rm"      "age"
##  [8] "dis"     "rad"     "tax"     "ptratio" "lstat"   "medv"
```

a) Use the poly() function to fit a cubic polynomial regression to predict nox using dis. Report the regression output, and plot the resulting data and polynomial fits.

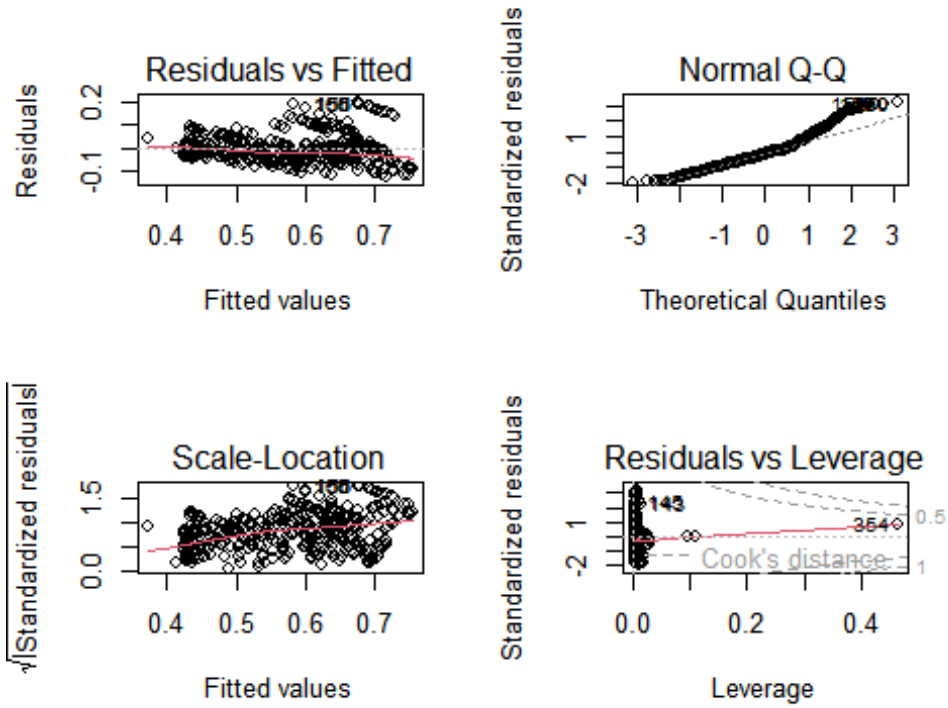
```
plot(Boston$dis, Boston$nox, xlab="dis", ylab="nox")
```




```
poly3_fit <- lm(nox ~ poly(dis,3,raw=TRUE) , data = Boston)
summary(poly3_fit)

##
## Call:
## lm(formula = nox ~ poly(dis, 3, raw = TRUE), data = Boston)
##
## Residuals:
##       Min        1Q    Median        3Q       Max
## -0.121130 -0.040619 -0.009738  0.023385  0.194904
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 0.9341281  0.0207076  45.110  < 2e-16 ***
## poly(dis, 3, raw = TRUE)1 -0.1820817  0.0146973 -12.389  < 2e-16 ***
## poly(dis, 3, raw = TRUE)2  0.0219277  0.0029329   7.476 3.43e-13 ***
## poly(dis, 3, raw = TRUE)3 -0.0008850  0.0001727  -5.124 4.27e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06207 on 502 degrees of freedom
## Multiple R-squared:  0.7148, Adjusted R-squared:  0.7131
## F-statistic: 419.3 on 3 and 502 DF,  p-value: < 2.2e-16

par(mfrow=c(2,2))
plot(poly3_fit)
```



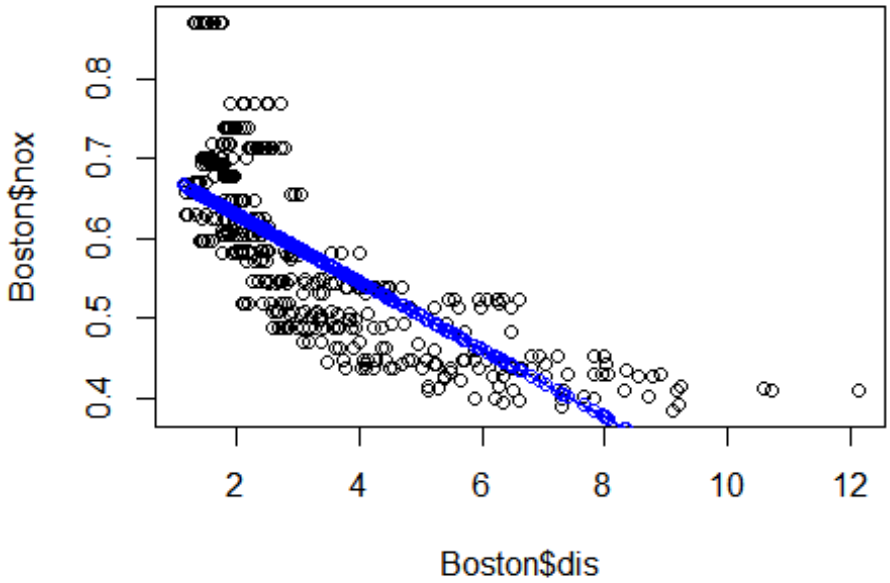

```
res <- sum(residuals.lm(poly3_fit)^2)
res

## [1] 1.934107
```

As we can see on the first plot, the two variables dis and nox seems to have a non-linear relationship. From the cubic polynomial fit, all the three polynomial degrees are significant predictors when fitting a cubic polynomial model to predict nox in the Boston data set. The residuals seems to be normally distributed and no relevant patterns. The associated RSS is low, 1.93.

b) Plot the polynomial fits for a range of different polynomial degrees (say, from 1 to 10), and report the associated residual sum of squares.

```
fit_resid = rep(NA,10)
for (i in 1:10){
  degree = i
  poly_fit <- lm(nox ~ poly(dis, i ,raw=TRUE) , data = Boston)
  fit_resid[i] = sum(residuals.lm(poly_fit)^2)
  plot(Boston$dis, Boston$nox)
  lines(Boston$dis, poly_fit$fitted.values, type="b", col ="blue")
}
```


Boston$nox
0.4
0.5
0.6
0.7
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2
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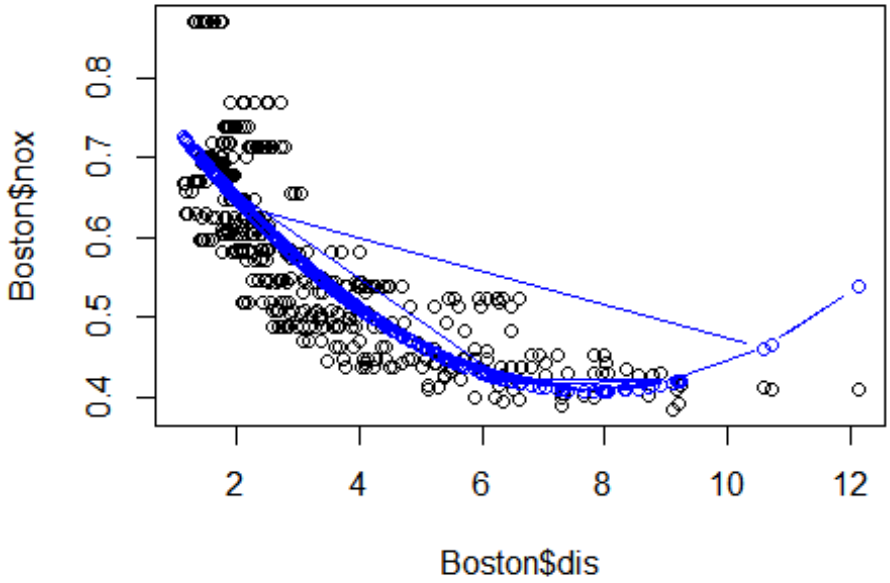
8
10
12
Boston$dis


Boston$nox
0.4
0.5
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0.7
0.8
2
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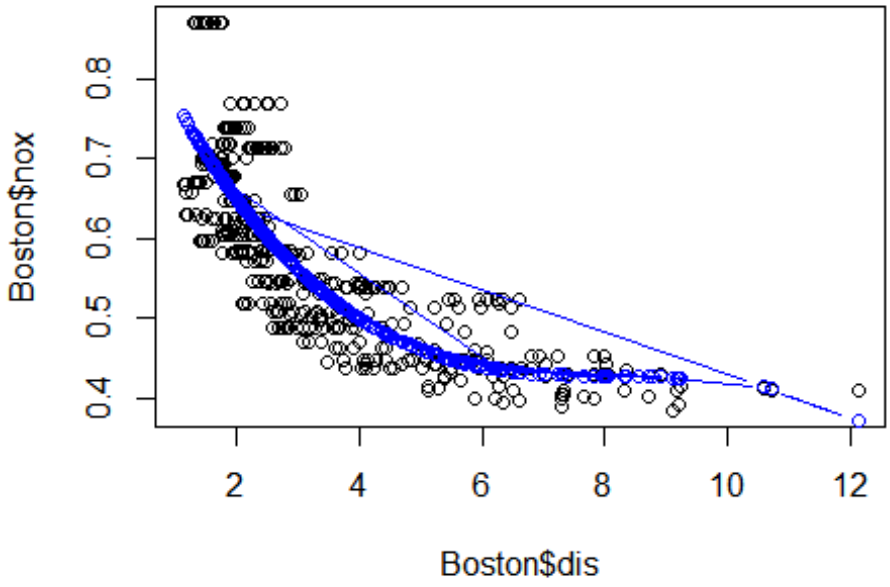
10
12
Boston$dis


Boston$nox
0.4
0.5
0.6
0.7
0.8
2
4
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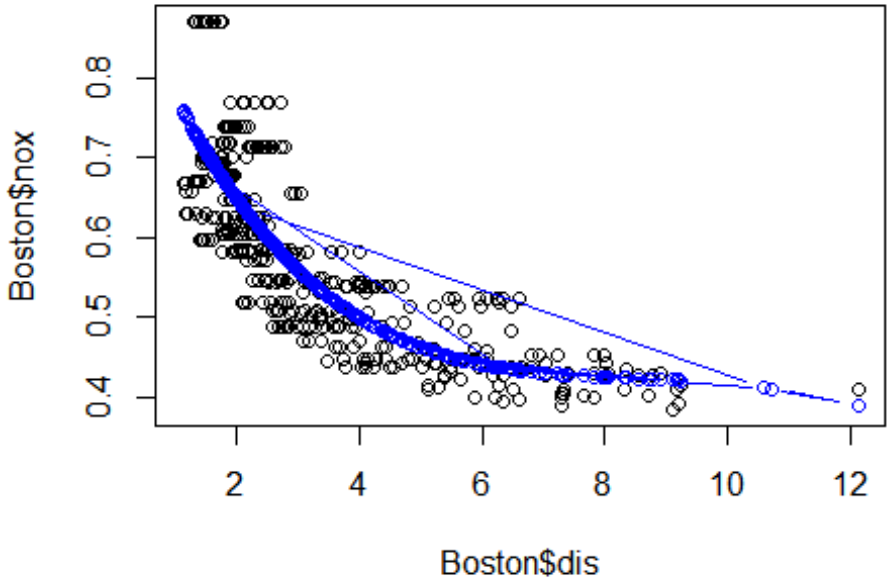
10
12
Boston$dis


Boston$nox
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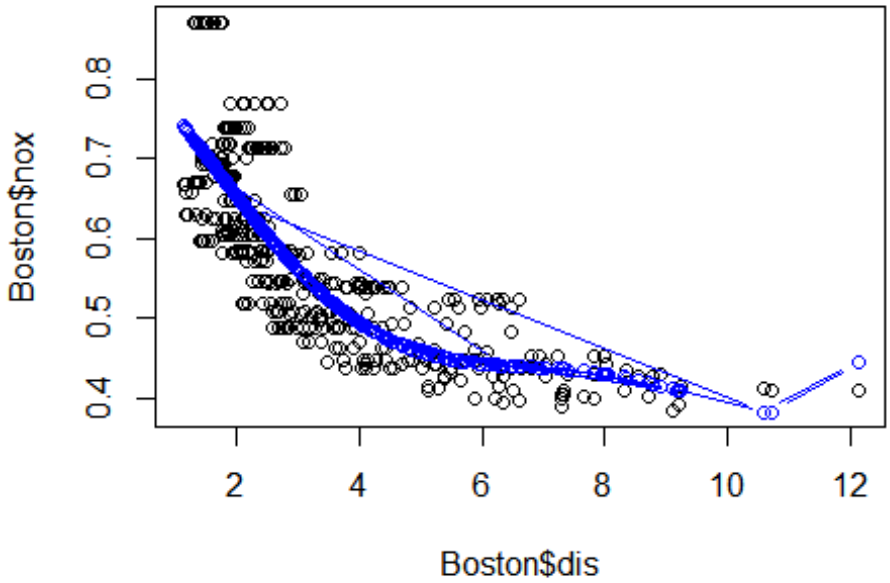
12
Boston$dis


Boston$nox
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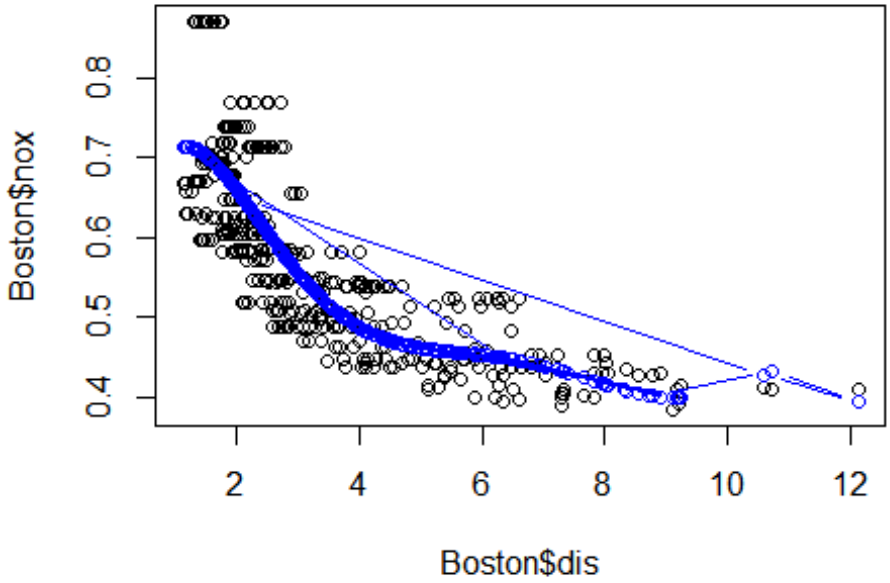
12
Boston$dis


Boston$nox
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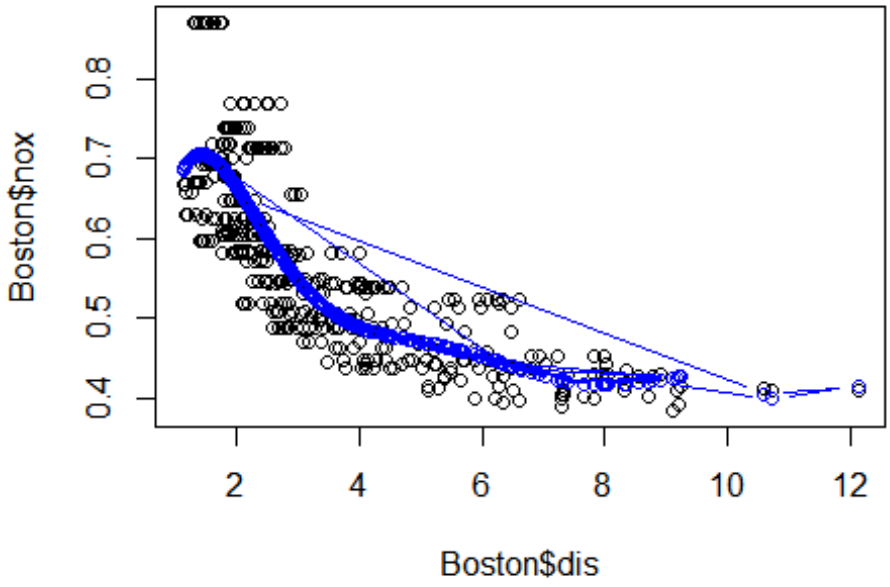
Boston$dis


Boston$nox
0.4
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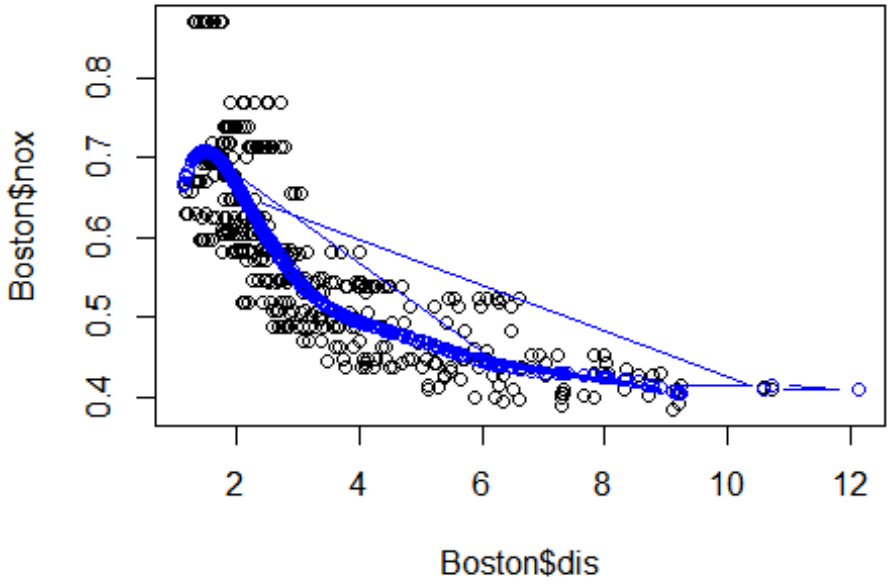
0.6
0.7
0.8
2
4
6
8
10
12
Boston$dis


Boston$nox
0.4
0.5
0.6
0.7
0.8
2
4
6
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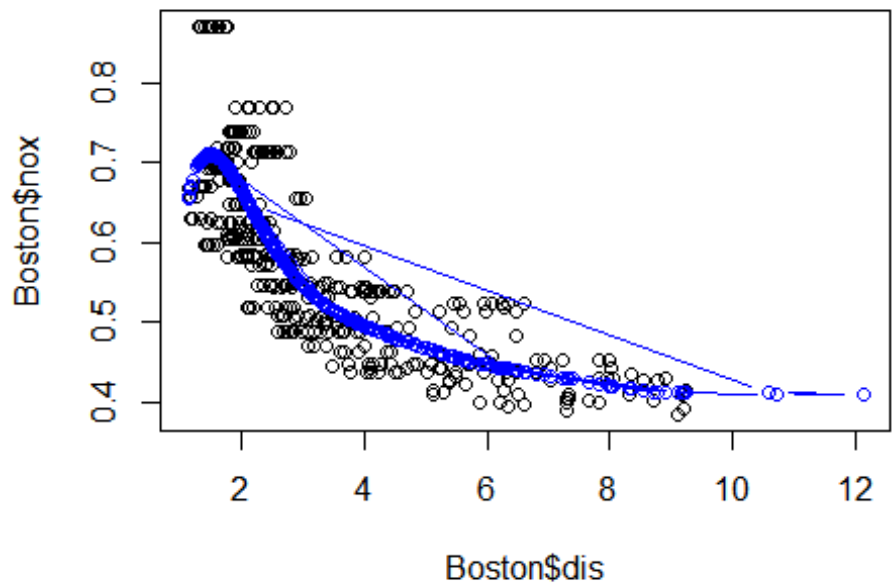
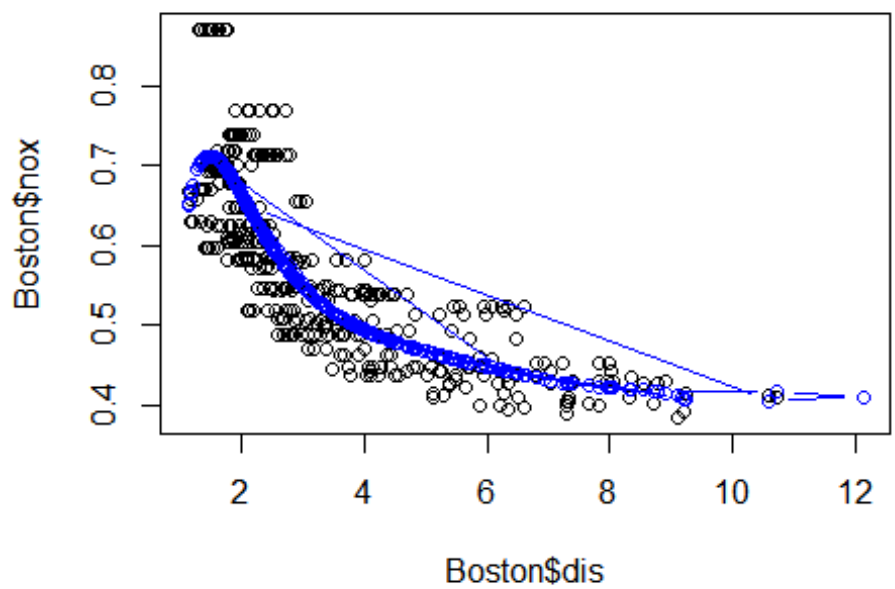
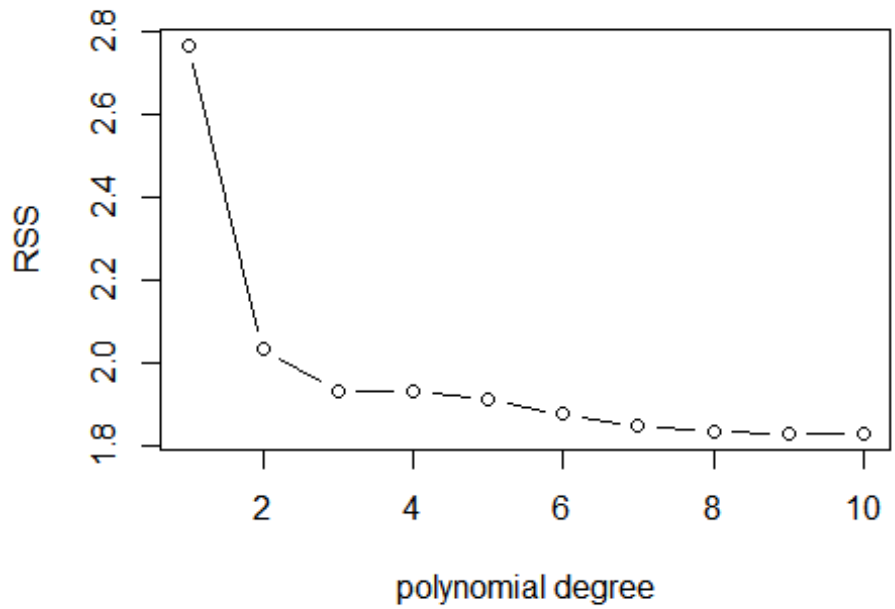
10
12
Boston$dis






```
df_resid <- data.frame(degree = 1:10, RSS = fit_resid)
par(mfrow=c(1,1))
lines(plot(df_resid , xlab="polynomial degree", type="b"))
```



When fitting a polynomial model in a range of degrees from 1 to 10, we can see that the associated RSS to each polynomial degree decreases. When we fit a linear model (degree=1), the RSS is the highest one as these two variables clearly does not have a linear relationship.

c) Perform some approach to select the optimal degree for the polynomial, and explain your results.

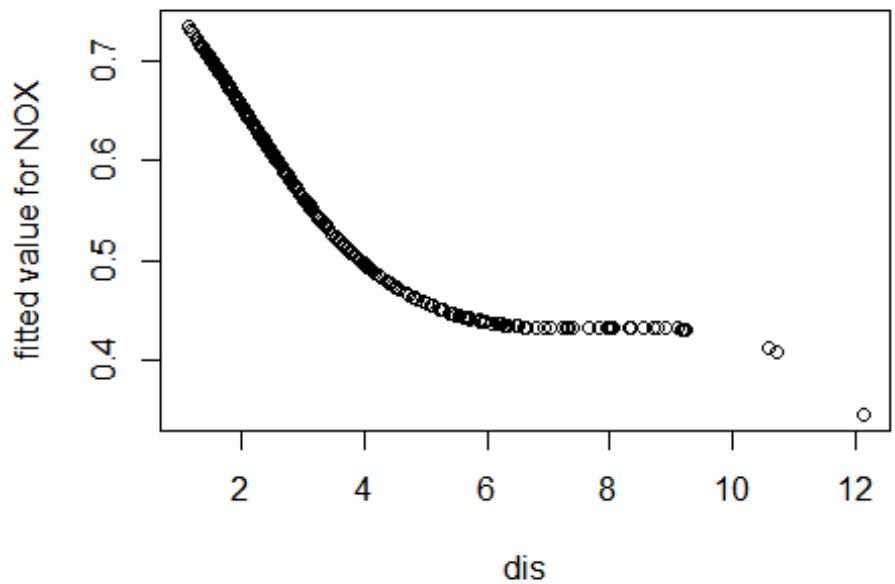
```
which.min(df_resid$RSS)

## [1] 10
```

As expected, the polynomial degree of 10 is the one that gives the lowest RSS and hence, is the one that best fits the data.

d) Use the bs() function to fit a regression spline to predict nox using dis. Report the output for the fit using four degrees of freedom. How did you choose the knots? Plot the resulting fit.

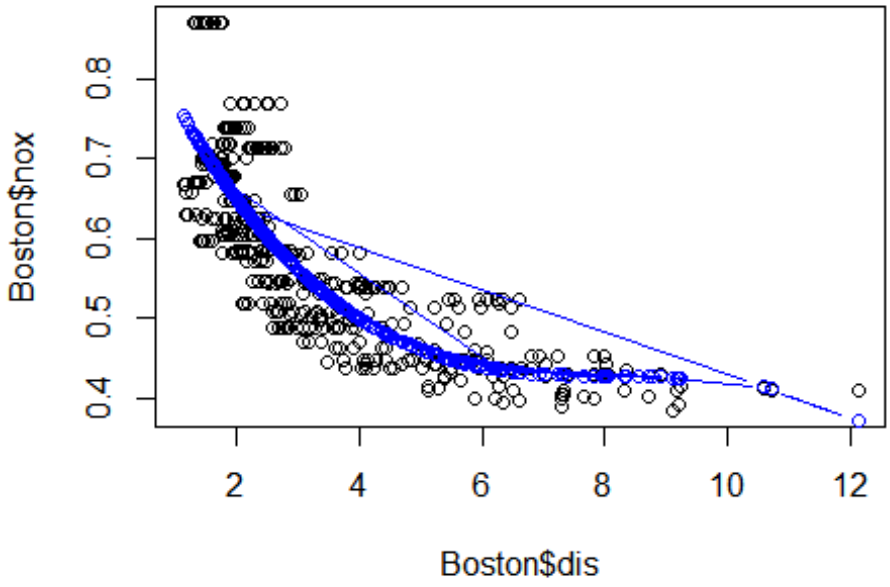
```
library(splines)
fit_spline <- lm(nox ~ bs(dis , df=4), data = Boston)
knot <- median(Boston$dis)
par(mfrow=c(1,1))
plot(Boston$dis, fit_spline$fitted.values, xlab="dis", ylab="fitted value for
NOX")
lines(x=knot, type="l")
```



A regression spline with 4 degrees of freedom is a quadratic polynomial with one knot (intercept + x + x^2 + knot). In this case, the knot was chosen as the median of values of "dis" in the data set (3.2). We can see the fitted values obtained in the spline on the above plot.

(e) Now fit a regression spline for a range of degrees of freedom, and plot the resulting fits and report the resulting RSS. Describe the results obtained.

```
spfit_resid = rep(NA,10)
for (i in 3:12){
  degree = i
  sp_fit <- lm(nox ~ bs(dis , df=i), data = Boston)
  spfit_resid[i] = sum(residuals.lm(sp_fit)^2)
  plot(Boston$dis, Boston$nox)
  lines(Boston$dis, sp_fit$fitted.values, type="b", col ="blue")
}
```


Boston$nox
0.4
0.5
0.6
0.7
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2
4
6
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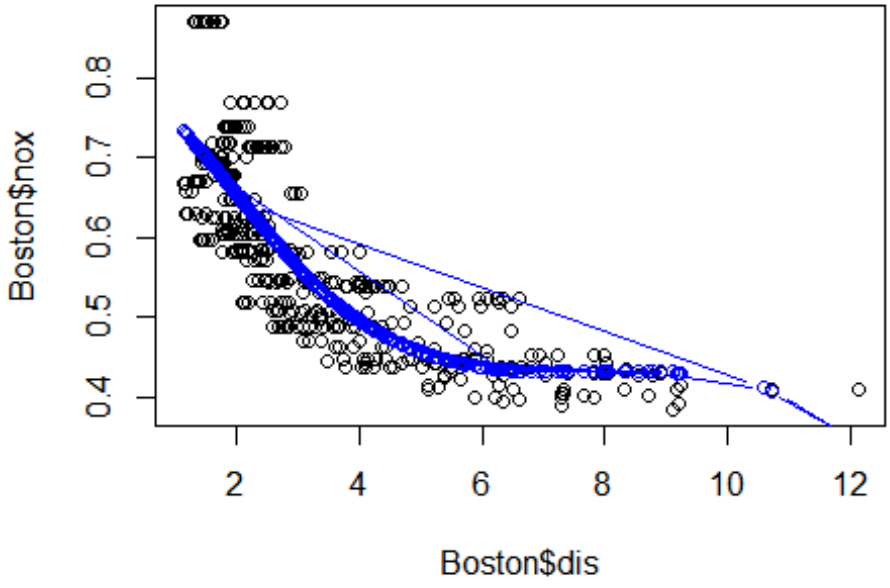
10
12
Boston$dis


Boston$nox
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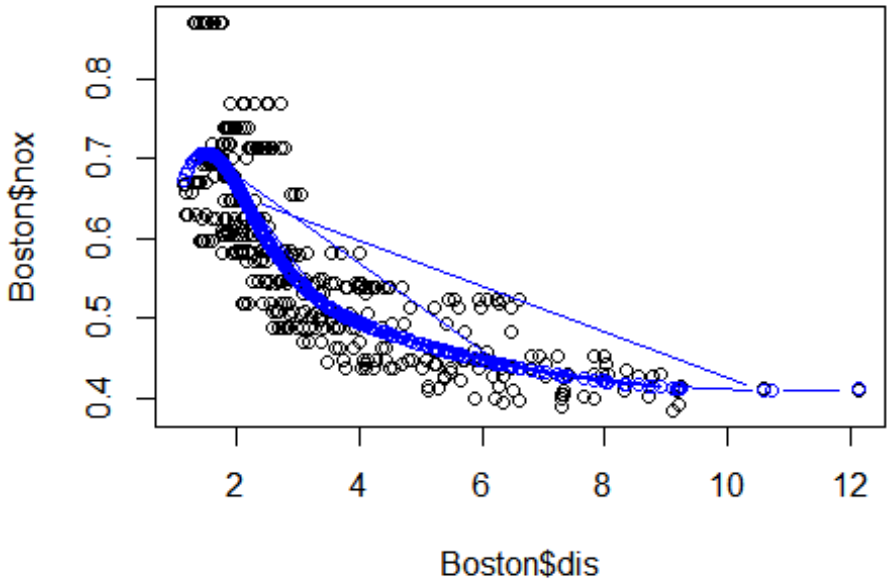
Boston$dis


Boston$nox
0.4
0.5
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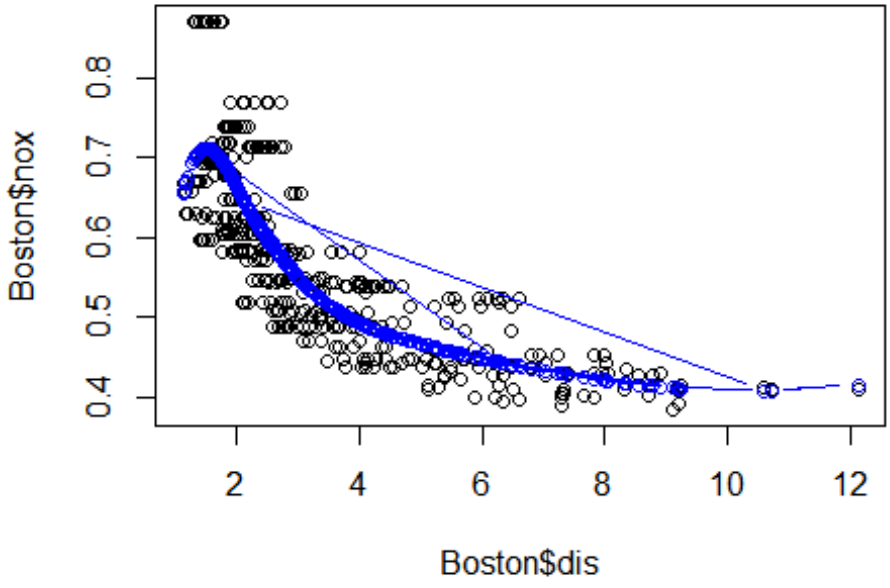
0.7
0.8
2
4
6
8
10
12
Boston$dis


Boston$nox
0.4
0.5
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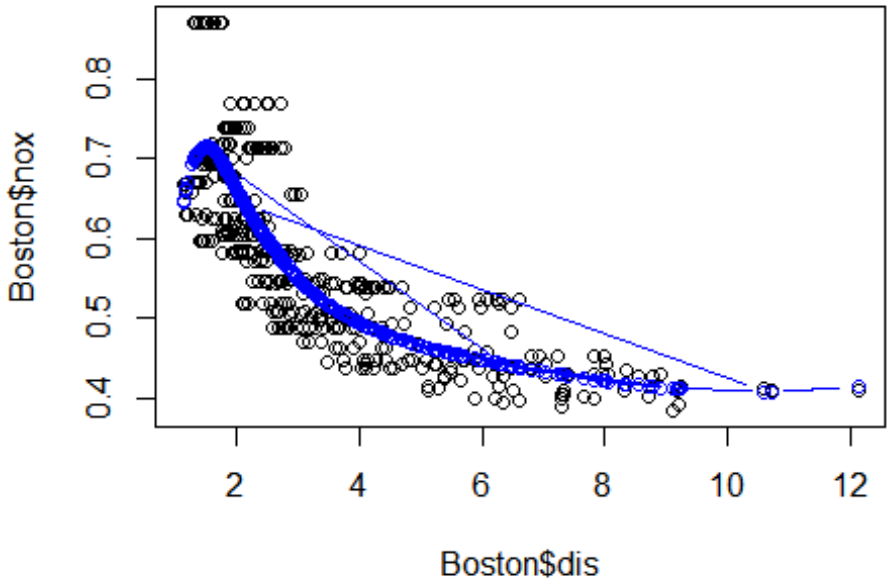
Boston$dis


Boston$nox
0.4
0.5
0.6
0.7
0.8
2
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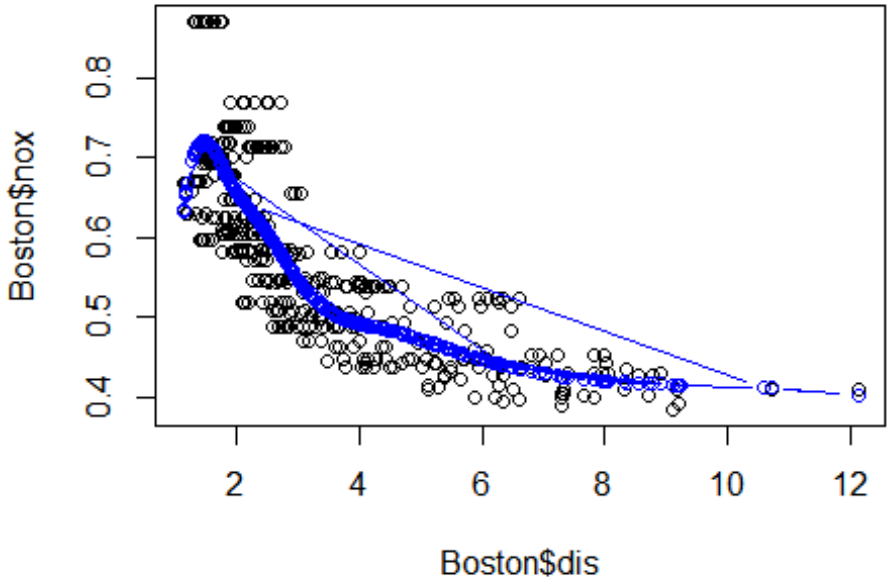
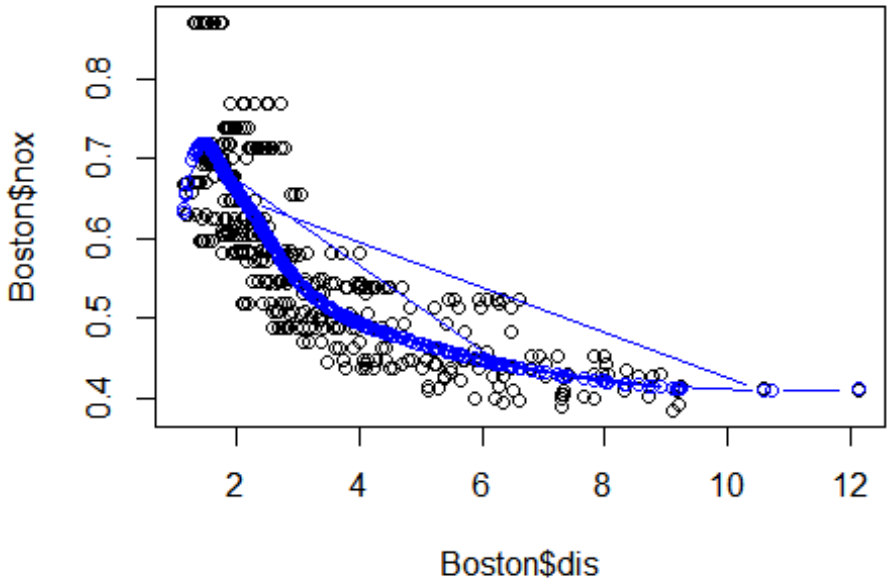
10
12
Boston$dis


Boston$nox
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
Boston$dis


Boston$nox
0.4
0.5
0.6
0.7
0.8
2
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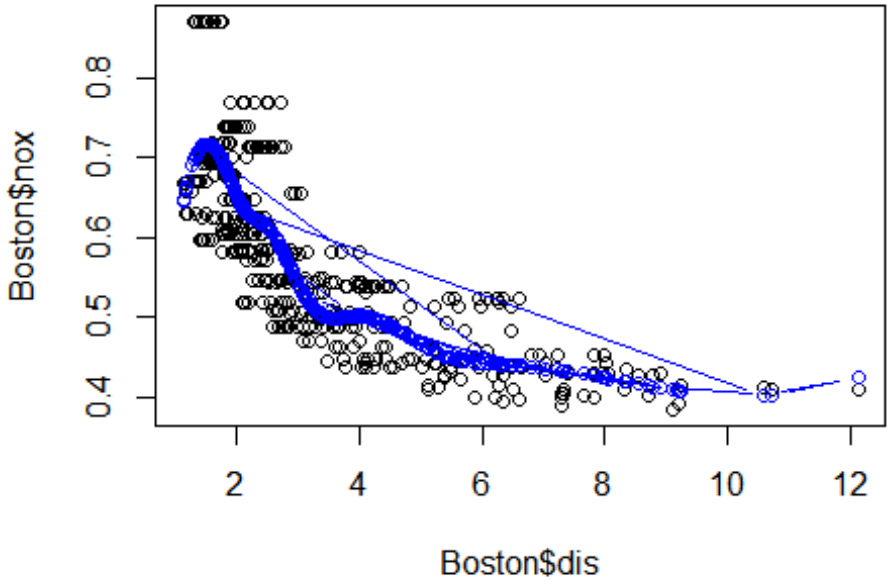
6
8
10
12
Boston$dis


Boston$nox
0.4
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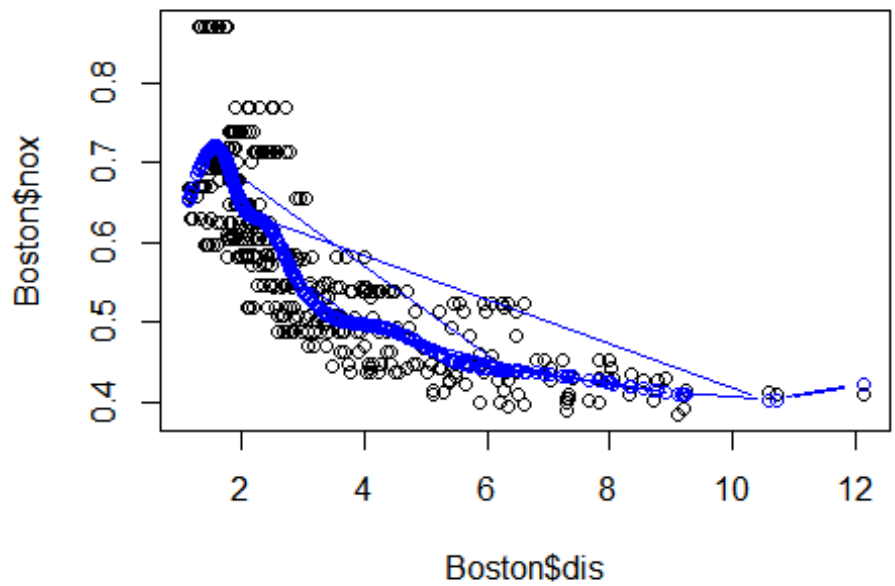
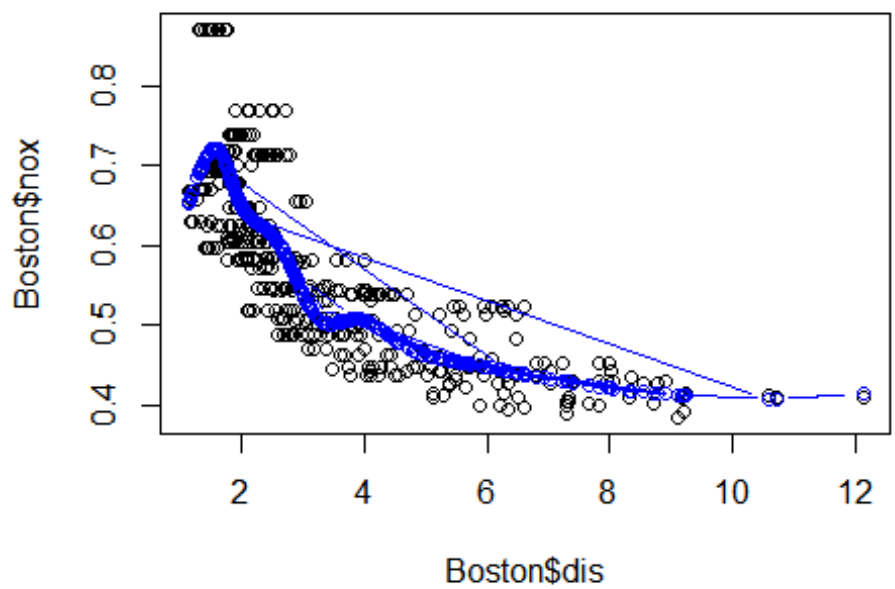
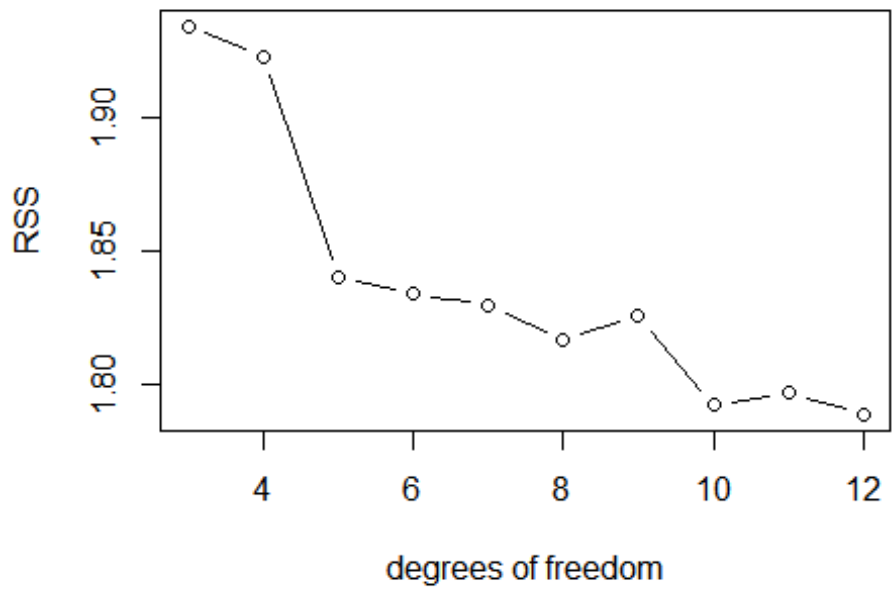
12
Boston$dis






```
df2_resid <- data.frame(df = 1:12, RSS = spfit_resid)
par(mfrow=c(1,1))
lines(plot(df2_resid[3:12,] , xlab="degrees of freedom", type="b"))
```



From all the regression splines fitted, we can see that the RSS keeps going down while having more degrees of freedom (this means, having more knots).

(f) Perform some approach in order to select the best degrees of freedom for a regression spline on this data. Describe your results.

```
which.min(df2_resid$RSS)

## [1] 12
```

The regression spline with the lowest RSS is the one with 12 degrees of freedom. This means, having a cubic spline with 8 knots for the splines (intercept, x, x^2, x^3, and the 8 knots)

Problem 10

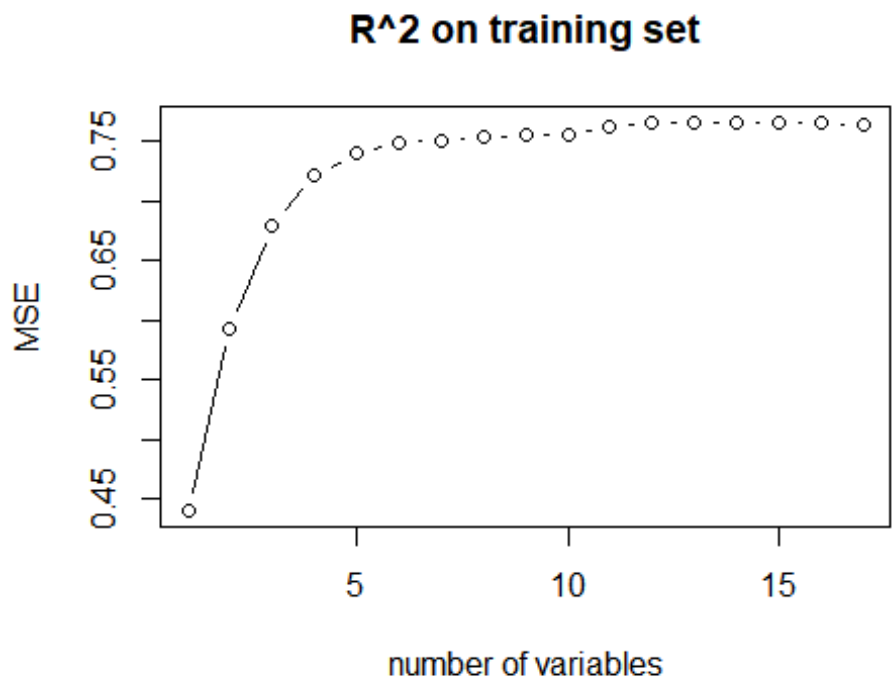
This question relates to the College data set.

```
data("College")
names(College)

##  [1] "Private"     "Apps"        "Accept"      "Enroll"      "Top10perc"
##  [6] "Top25perc"   "F.Undergrad" "P.Undergrad" "Outstate"    "Room.Board"
## [11] "Books"       "Personal"    "PhD"         "Terminal"    "S.F.Ratio"
## [16] "perc.alumni" "Expend"      "Grad.Rate"
```

a) Split the data into a training set and a test set. Using out-of-state tuition as the response and the other variables as the predictors, perform forward stepwise

selection on the training set in order to identify a satisfactory model that uses just a subset of the predictors.

```
library(leaps)
set.seed(1)
train <- sample(1:nrow(College),nrow(College)*2/3)
test <- (-train)
College.train <- College[train,]
College.test <-  College[test,]
fw_fit <- regsubsets(Outstate ~ . , data=College.train, method = "forward",
nvmax=17)
fw_fit_sum <- summary(fw_fit)
best_model <- which.max(fw_fit_sum$adjr2)
best_model

## [1] 13

coef(fw_fit,best_model)

##   (Intercept)    PrivateYes          Apps        Accept     Top10perc
## -1646.4323441  2246.8796422    -0.3611338     0.8167646    44.5415347
##     Top25perc   F.Undergrad    Room.Board      Personal      Terminal
##   -12.6489819    -0.1550593     0.9804990    -0.3686231    37.3053333
##     S.F.Ratio   perc.alumni        Expend     Grad.Rate
##   -56.1892591    47.3743338     0.1650308    20.1462005

fw_fit_rsq <- fw_fit_sum$adjr2
par(mfrow = c(1,1))
plot(fw_fit_rsq , main = "R^2 on training set" , xlab = "number of variables"
, ylab = "MSE", type = "b")
```




```
which.max(fw_fit_rsq)

## [1] 13
```

When performing a forward stepwise selection on the training data, we get the best model based on the highest R-squared, which means that the response is explained by the predictors in a higher percentage. This model is the one that uses 13 predictors out of the 15. The predictors and its coefficients are shown also.

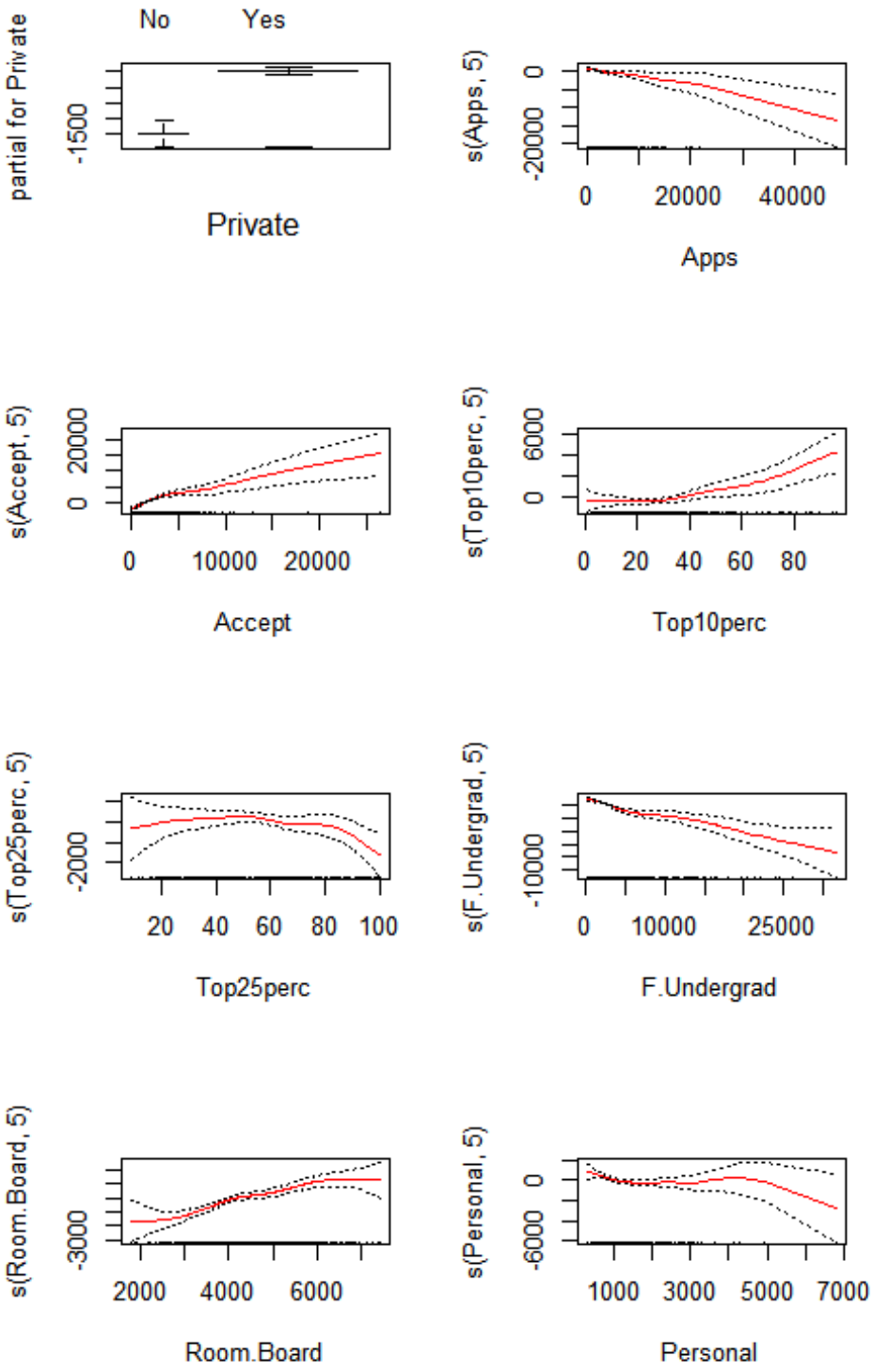
(b) Fit a GAM on the training data, using out-of-state tuition as the response and the features selected in the previous step as the predictors. Plot the results, and explain your findings.

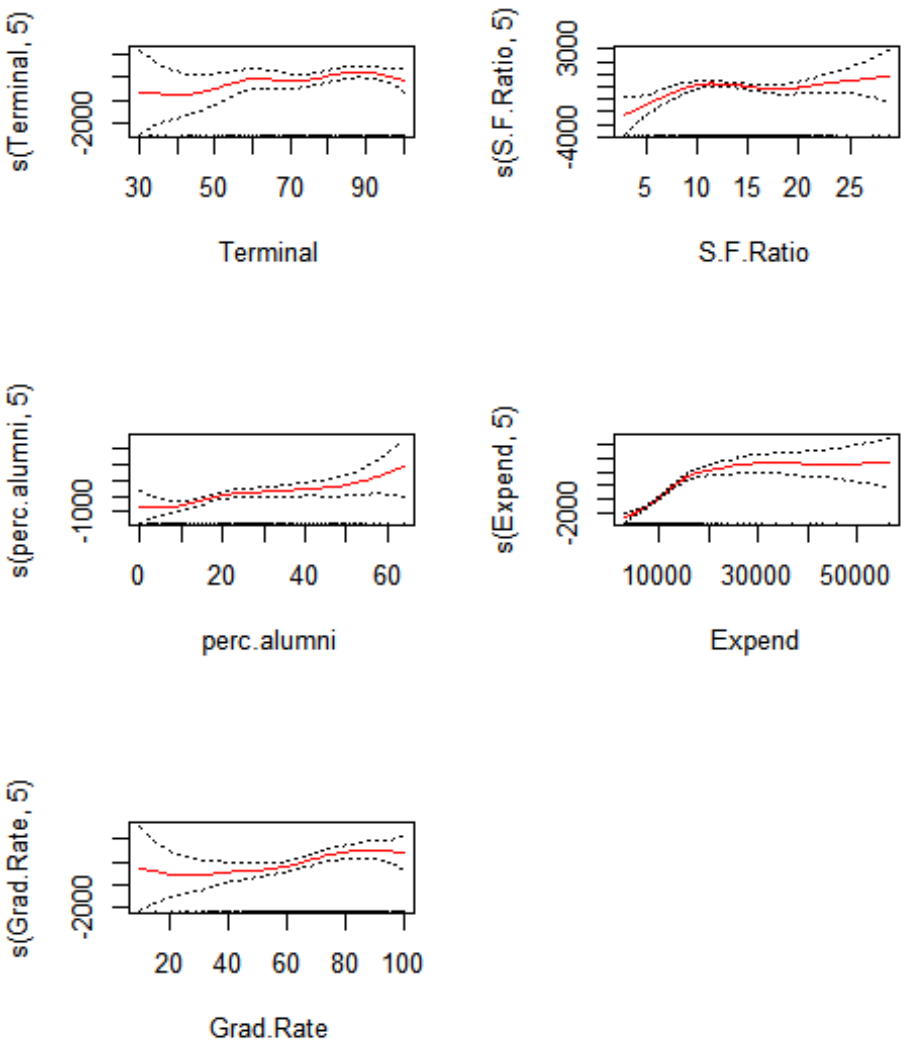
```
library(gam)

## Warning: package 'gam' was built under R version 4.2.3

## Loading required package: foreach

## Loaded gam 1.22-2

library(splines)
gam_fit1 <- gam(Outstate ~ Private + Apps + Accept + Top10perc + Top25perc +
F.Undergrad + Room.Board + Personal + Terminal + S.F.Ratio + perc.alumni +
Expend + Grad.Rate, data=College.train )
gam_fit2 <- gam(Outstate ~ Private + s(Apps,5) + s(Accept,5) + s(Top10perc,5)
+ s(Top25perc,5) + s(F.Undergrad,5) + s(Room.Board,5) + s(Personal,5) +
s(Terminal,5) + s(S.F.Ratio,5) + s(perc.alumni,5) + s(Expend,5) +
```

```r
s(Grad.Rate,5), data=College.train )
anova(gam_fit1 , gam_fit2 , test="F")
```

```
## Analysis of Deviance Table
##
## Model 1: Outstate ~ Private + Apps + Accept + Top10perc + Top25perc +
##     F.Undergrad + Room.Board + Personal + Terminal + S.F.Ratio +
##     perc.alumni + Expend + Grad.Rate
## Model 2: Outstate ~ Private + s(Apps, 5) + s(Accept, 5) + s(Top10perc,
##     5) + s(Top25perc, 5) + s(F.Undergrad, 5) + s(Room.Board,
##     5) + s(Personal, 5) + s(Terminal, 5) + s(S.F.Ratio, 5) +
##     s(perc.alumni, 5) + s(Expend, 5) + s(Grad.Rate, 5)
##   Resid. Df Resid. Dev Df  Deviance      F    Pr(>F)
## 1       504 2068933122
## 2       456 1422315144 48 646617978 4.3189 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
par(mfrow = c(2, 2))
plot(gam_fit2, se = TRUE , col = "red")
```


partial for Private
No
Yes
-1500
Private
s(Apps, 5)
0
-20000
0
20000
40000
Apps
s(Accept, 5)
0
20000
0
10000
20000
Accept
s(Top10perc, 5)
0
6000
0
20
40
60
80
Top10perc
s(Top25perc, 5)
-2000
20
40
60
80
100
Top25perc
s(F.Undergrad, 5)
-10000
0
10000
25000
F.Undergrad
s(Room.Board, 5)
-3000
2000
4000
6000
Room.Board
s(Personal, 5)
0
-6000
1000
3000
5000
7000
Personal



Compared are two GAM models, one fitting the predictors as linear and the second one using basis functions with 5 degrees of freedom. on each of the quantitative predictors. When comparing both on the ANOVA, we can see that the p-value is very small (2.2e-16) which concludes that using

basis functions is a better model than the first one. There it is also included the associated plots to this GAM fit.

(c) Evaluate the model obtained on the test set, and explain the results obtained.

```
College_pred2 <- predict.Gam(gam_fit2 , newdata = College.test)
gam_MSE2 <- mean((College_pred2 - College.test$Outstate)^2)
gam_MSE2

## [1] 3229970
```

Even though the GAM2 model fits better than the GAM1, the MSE on the testing data is too high so we can conclude that this model is not good at generalizing, as the variance is too high.

(d) For which variables, if any, is there evidence of a non-linear relationship with the response?

```
summary(gam_fit2)

##
## Call: gam(formula = Outstate ~ Private + s(Apps, 5) + s(Accept, 5) +
##     s(Top10perc, 5) + s(Top25perc, 5) + s(F.Undergrad, 5) + s(Room.Board,
##     5) + s(Personal, 5) + s(Terminal, 5) + s(S.F.Ratio, 5) +
##     s(perc.alumni, 5) + s(Expend, 5) + s(Grad.Rate, 5), data =
College.train)
## Deviance Residuals:
##       Min        1Q    Median        3Q       Max
## -6059.322 -1070.578     0.925  1063.952  6979.605
##
## (Dispersion Parameter for gaussian family taken to be 3119111)
##
##     Null Deviance: 9046608321 on 517 degrees of freedom
## Residual Deviance: 1422315144 on 456.0001 degrees of freedom
## AIC: 9275.669
##
## Number of Local Scoring Iterations: NA
##
## Anova for Parametric Effects
##                     Df     Sum Sq    Mean Sq  F value    Pr(>F)
## Private              1 2455186324 2455186324 787.1429 < 2.2e-16 ***
## s(Apps, 5)           1  992328372  992328372 318.1446 < 2.2e-16 ***
## s(Accept, 5)         1  122745539  122745539  39.3527 8.228e-10 ***
## s(Top10perc, 5)      1 1103461375 1103461375 353.7743 < 2.2e-16 ***
## s(Top25perc, 5)      1    6152677    6152677   1.9726  0.160856
## s(F.Undergrad, 5)    1  291864244  291864244  93.5729 < 2.2e-16 ***
## s(Room.Board, 5)     1  570196980  570196980 182.8075 < 2.2e-16 ***
## s(Personal, 5)       1   52025024   52025024  16.6794 5.228e-05 ***
## s(Terminal, 5)       1   80030322   80030322  25.6581 5.930e-07 ***
## s(S.F.Ratio, 5)      1   91883068   91883068  29.4581 9.300e-08 ***
## s(perc.alumni, 5)    1  114527858  114527858  36.7181 2.861e-09 ***
## s(Expend, 5)         1  478681405  478681405 153.4672 < 2.2e-16 ***
```

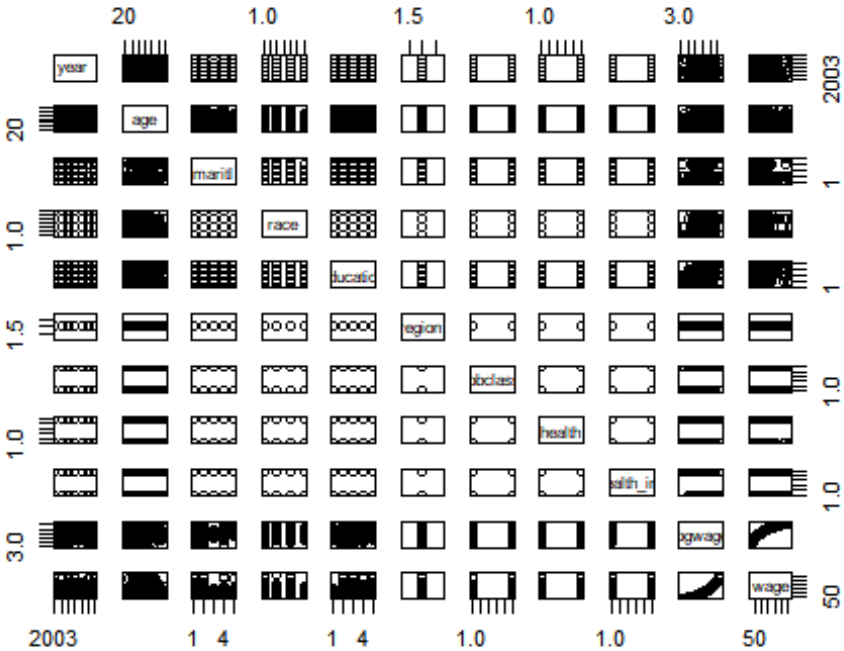
```
## s(Grad.Rate, 5)     1   33512833   33512833  10.7444  0.001126 **
## Residuals         456 1422315144    3119111
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Anova for Nonparametric Effects
##                   Npar Df  Npar F     Pr(F)
## (Intercept)
## Private
## s(Apps, 5)              4  2.2438  0.063453 .
## s(Accept, 5)            4 12.1695 2.081e-09 ***
## s(Top10perc, 5)         4  3.1349  0.014628 *
## s(Top25perc, 5)         4  2.5813  0.036689 *
## s(F.Undergrad, 5)       4  3.0004  0.018327 *
## s(Room.Board, 5)        4  2.0523  0.086093 .
## s(Personal, 5)          4  2.3177  0.056341 .
## s(Terminal, 5)          4  1.4504  0.216379
## s(S.F.Ratio, 5)         4  4.1846  0.002445 **
## s(perc.alumni, 5)       4  1.2003  0.309924
## s(Expend, 5)            4 21.9834 < 2.2e-16 ***
## s(Grad.Rate, 5)         4  0.9456  0.437280
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

From the Anova for Nonparametric Effects we can detect the predictors that show evidence of a non-linear relationship with the response (marked with *): Accept, Top10perc, Top25perc, F.Ratio, and Expend.

Problem 7

The Wage data set contains a number of other features not explored in this chapter, such as marital status (maritl), job class (jobclass), and others. Explore the relationships between some of these other predictors and wage, and use non-linear fitting techniques in order to fit flexible models to the data. Create plots of the results obtained, and write a summary of your findings.

```
data("Wage")
names(Wage)

##  [1] "year"       "age"        "maritl"     "race"       "education"
##  [6] "region"     "jobclass"   "health"     "health_ins" "logwage"
## [11] "wage"

plot(Wage)
```

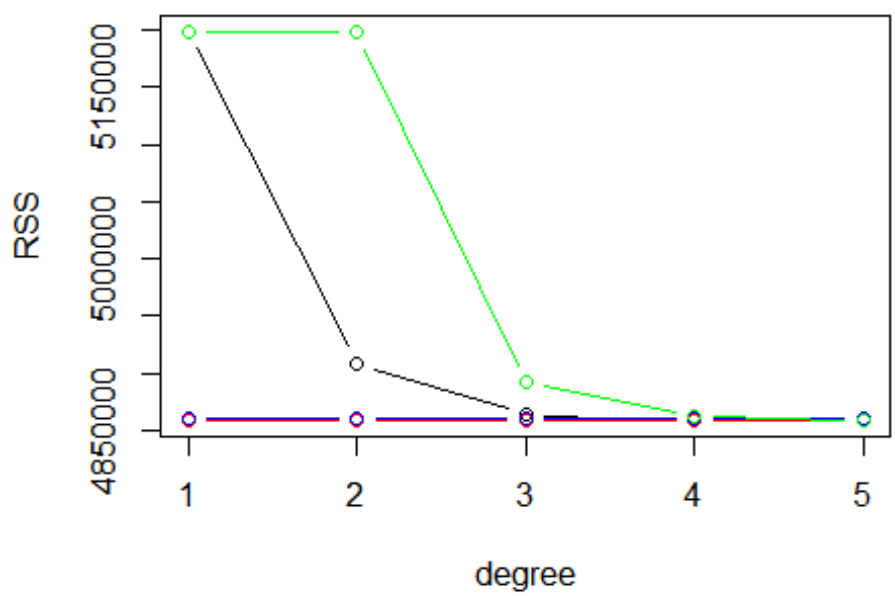


There seems to be non-linear relationships with these predictors: maritl, race, education. We will be exploring two of these, maritl and education with different techniques.

- for maritl

```
resid_poly_fit1 = rep(NA,5)
resid_cubic_spline_fit1 = rep(NA,5)
resid_nat_spline_fit1 = rep(NA,5)
resid_loc_reg_fit1 = rep(NA,5)
resid_sm_spline_fit1 = rep(NA,5)
d <- c(1.01,2,3,4,5)
e <- c(0.5, 0.55, 0.65, 0.70, 0.80)
for (i in 1:5){
  degree = i
  poly_fit1 <- lm(wage ~ poly(maritl, i ,raw=TRUE) , data = Wage)
  resid_poly_fit1[i] = sum(residuals.lm(poly_fit1)^2)

  cubic_spline_fit1 <- lm(wage ~ bs(as.double(Wage$maritl) , df=3+i), data =
Wage)
  resid_cubic_spline_fit1[i] = sum((residuals.lm(cubic_spline_fit1))^2)

  nat_spline_fit1 <- lm(wage ~ ns(as.double(Wage$maritl) , df=3+i), data =
Wage)
  resid_nat_spline_fit1[i] = sum((residuals.lm(nat_spline_fit1))^2)

  sm_spline_fit1 <- smooth.spline(as.double(Wage$maritl) , Wage$wage ,
df=d[i] , tol = 1e-6)
  resid_sm_spline_fit1[i] <- sum((residuals(sm_spline_fit1))^2)
```

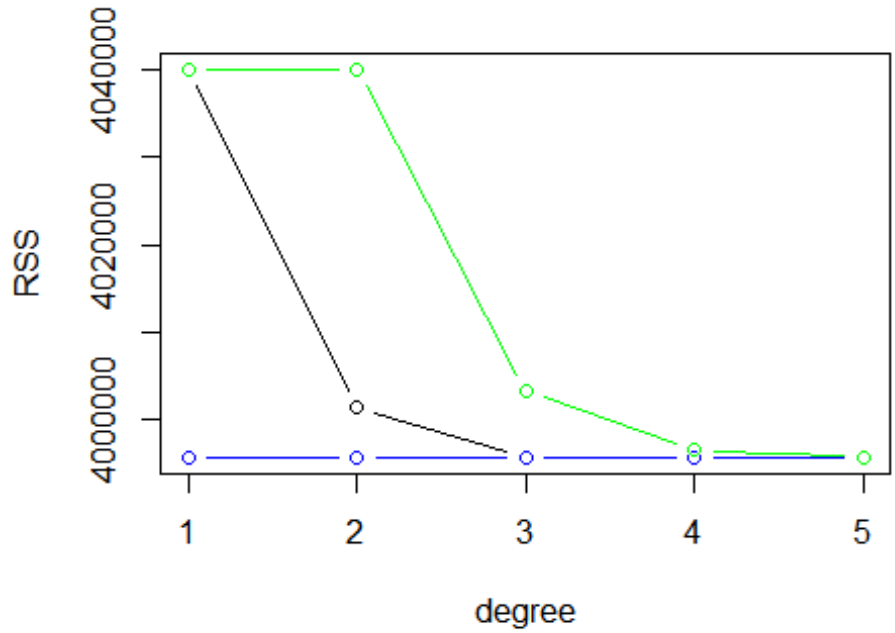
```
}


df_resid_fit1 <- data.frame(degree = 1:5, RSS_poly = resid_poly_fit1,
RSS_cubic_sp = resid_cubic_spline_fit1, RSS_nat_spline =
resid_nat_spline_fit1, RSS_sm_spline =  resid_sm_spline_fit1)
df_resid_fit1

##   degree RSS_poly RSS_cubic_sp RSS_nat_spline RSS_sm_spline
## 1      1  5198391      4858941        4860060       5198391
## 2      2  4908683      4858941        4860060       5198391
## 3      3  4863634      4858941        4860060       4892096
## 4      4  4858941      4858941        4860060       4862791
## 5      5  4858941      4858941        4860060       4858941

par(mfrow=c(1,1))
plot(df_resid_fit1$RSS_poly , xlab="degree", ylab="RSS" , type="b")
lines(df_resid_fit1$RSS_cubic_sp, col="red", type="b")
lines(df_resid_fit1$RSS_nat_spline, col="blue", type="b")
lines(df_resid_fit1$RSS_sm_spline, col="green", type="b")
```



In the table and plot above we can see the RSS associated with each of the non-linear techniques explored using marital status (maritl) as the predictor. For some of the methods, it does not show a relevant change when using different degrees of freedom (as for the cubic splines and natural splines). However, the polynomial fit and the smoothing splines do have an improvement on the RSS when adding more degrees of freedom.

- for education

```
resid_poly_fit2 = rep(NA,5)
resid_cubic_spline_fit2 = rep(NA,5)
resid_nat_spline_fit2 = rep(NA,5)
resid_loc_reg_fit2 = rep(NA,5)
resid_sm_spline_fit2 = rep(NA,5)
d <- c(1.01,2,3,4,5)
e <- c(0.5, 0.55, 0.65, 0.70, 0.80)
for (i in 1:5){
  degree = i
  poly_fit2 <- lm(wage ~ poly(education, i ,raw=TRUE) , data = Wage)
  resid_poly_fit2[i] = sum(residuals.lm(poly_fit2)^2)

  cubic_spline_fit2 <- lm(wage ~ bs(as.double(Wage$education) , df=3+i), data
= Wage)
  resid_cubic_spline_fit2[i] = sum((residuals.lm(cubic_spline_fit2))^2)

  nat_spline_fit2 <- lm(wage ~ ns(as.double(Wage$education) , df=2+i), data =
Wage)
  resid_nat_spline_fit2[i] = sum((residuals.lm(nat_spline_fit2))^2)

  sm_spline_fit2 <- smooth.spline(as.double(Wage$education) , Wage$wage ,
df=d[i] , tol = 1e-6)
  resid_sm_spline_fit2[i] <- sum((residuals(sm_spline_fit2))^2)
}


df_resid_fit2 <- data.frame(degree = 1:5, RSS_poly = resid_poly_fit2,
RSS_cubic_sp = resid_cubic_spline_fit2, RSS_nat_spline =
resid_nat_spline_fit2, RSS_sm_spline =  resid_sm_spline_fit2)
par(mfrow=c(1,1))
df_resid_fit2

##   degree RSS_poly RSS_cubic_sp RSS_nat_spline RSS_sm_spline
## 1      1  4040005      3995721        3995784       4040005
## 2      2  4001380      3995721        3995721       4040005
## 3      3  3995726      3995721        3995721       4003274
## 4      4  3995721      3995721        3995721       3996530
## 5      5  3995721      3995721        3995721       3995721

plot(df_resid_fit2$RSS_poly , xlab="degree", ylab="RSS" , type="b")
lines(df_resid_fit2$RSS_cubic_sp, col="red", type="b")
lines(df_resid_fit2$RSS_nat_spline, col="blue", type="b")
lines(df_resid_fit2$RSS_sm_spline, col="green", type="b")
```



In the table and plot above we can see the RSS associated with each of the non-linear techniques explored using education as the predictor. For some of the methods, it does not show a relevant change when using different degrees of freedom (for example in the cubic splines and natural splines). However, the polynomial fit and the smoothing splines do have an improvement on the RSS when adding more degrees of freedom.

HOMEWORK #3 – Problem 1

① $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 h(x)$

$\hookrightarrow (x - z)^3$ if $x > z$

0 o.w

(a) $f_1(x) = a_1 + b_1 x + c_1 x^2 + d_1 x^3$

s.t. $f(x) = f_1(x)$ for all $x \leq z$

$a_1 = \beta_0 \quad b_1 = \beta_1 \quad c_1 = \beta_2 \quad d_1 = \beta_3$

(b) $f_2(x) = a_2 + b_2 x + c_2 x^2 + d_2 x^3$

s.t. $f(x) = f_2(x)$ for all $x > z$

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x^3 - 3x^2 z + 3xz^2 - z^3)$$
$$= \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^3 - 3\beta_4 z x^2 + 3\beta_4 z^2 x - \beta_4 z^3$$
$$= \underbrace{(\beta_0 - \beta_4 z^3)}_{a_2} + \underbrace{(\beta_1 + 3\beta_4 z^2)}_{b_2} x + \underbrace{(\beta_2 - 3\beta_4 z)}_{c_2} x^2 + \underbrace{(\beta_3 + \beta_4)}_{d_2} x^3$$

$a_2 = \beta_0 - \beta_4 z^3 \quad b_2 = \beta_1 + 3\beta_4 z^2 \quad c_2 = \beta_2 - 3\beta_4 z \quad d_2 = \beta_3 + \beta_4$

(c) $f_1(z) = \beta_0 + \beta_1 (z) + \beta_2 (z)^2 + \beta_3 (z)^3$

$$f_2(z) = \beta_0 - \beta_4 z^3 + (\beta_1 + 3\beta_4 z^2) z + (\beta_2 - 3\beta_4 z) z^2 + (\beta_3 + \beta_4) z^3$$
$$= \beta_0 - \cancel{\beta_4 z^3} + \beta_1 z + \cancel{3\beta_4 z^3} + \beta_2 z^2 - \cancel{3\beta_4 z^3} + \beta_3 z^3 + \cancel{\beta_4 z^3}$$
$$=$$

$$f_2(z) = \beta_0 + \beta_1 z + \beta_2 z^2 + \beta_3 z^3$$

$\Rightarrow f_1(z) = f_2(z) \rightarrow f(x)$ is continuous at z

(d) $f_1(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$

$f_1'(x) = \beta_1 + 2\beta_2 x + 3\beta_3 x^2$

$f_1'(z) = \beta_1 + 2\beta_2 z + 3\beta_3 z^2$

$f_2(x) = a_2 + b_2 x + c_2 x^2 + d_2 x^3$

$f_2'(x) = b_2 + 2c_2 x + 3d_2 x^2$

$f_2'(z) = \beta_1 + 3\beta_4 z^2 + 2(\beta_2 - 3\beta_4 z) z + 3(\beta_3 + \beta_4) z^2$

$f_2'(z) = \beta_1 + \cancel{3\beta_4 z^2} + 2\beta_2 z - \cancel{6\beta_4 z^2} + 3\beta_3 z^2 + \cancel{3\beta_4 z^2}$

$f_2'(z) = \beta_1 + 2\beta_2 z + 3\beta_3 z^2$

$\Rightarrow f_1'(z) = f_2'(z) \rightarrow f'(x)$ is continuous at z

(e) $f_1''(x) = 2\beta_2 + 6\beta_3 x$

$f_1''(z) = 2\beta_2 + 6\beta_3 z$

$f_2''(x) = 2c_2 + 6d_2 x$

$f_2''(z) = 2(\beta_2 - 3\beta_4 z) + 6(\beta_3 + \beta_4) z$

$= 2\beta_2 - \cancel{6\beta_4 z} + 6\beta_3 z + \cancel{6\beta_4 z}$

$f_2''(z) = 2\beta_2 + 6\beta_3 z$

$\Rightarrow f_1''(z) = f_2''(z) \rightarrow f''(x)$ is continuous at z